MILITARY SPECIFICATION

CODER-DECODER GROUP

OX-54/TYC-39(V)

This specification is approved for use by Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. - This specification establishes the performance and test requirements and qualifications of the air or vehicular transportable CODER-DECODER GROUP OX-54/TYC-39(V).

1.2 Description. - CODER-DECODER GROUP OX-54/TYC-39(V) hereinafter called the shelter facility, includes a modified S-280B/G shelter, and all racks, brackets, shelves and tie-downs, monitor panels, electrical components and wiring to enable installation and operation of the Inventory COMSEC Equipments listed in Sec 6.1.

2. APPLICABLE DOCUMENTS

2.1 Issue of Documents. - The following documents of the issue in effect on date of invitation for bids or request for proposal, forms part of this specification to the extent specified herein. In case of conflict between the document referenced and the contents of this specification, the contents of this specification shall be considered a superseding requirement.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Communications-Electronics Command, ATTN: DRSEL-PED-CM-DM, Ft. Monmouth, New Jersey 07703, by using the self-addressed Standardization of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Packaging, Methods of
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipments)
MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickle Alloys, Heat-Resisting Alloys Titanium, Spot and Seam
MIL-T-7928	Terminals, Lug and Splice, Crimp Style Copper
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-W-45205	Welding, Gas, Metal-Arc and Gas, Tungsten-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor
MIL-S-55507	Shelter, Electrical Equipment (with or without equipment), Packaging of

STANDARD

MILITARY

MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461A NOTICE 4	Electromagnetic Interference Characteristics; Requirements for equipment subsystems, and systems
MIL-STD-462, NOTICE 3	Electromagnetic Interference Characteristics, measurement of

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MILITARY (Continued)

MIL-STD-810C	Environmental Test Methods
MIL-STD-1474(MI) NACSEM 5100	Noise Limits for Army Material Compromising Emanations Laboratory Test Standard, Electromagnetics (Title UNCLASSIFIED, Document, CONFIDENTIAL).

DRAWINGS

SM-B-814704	Line Conditioning Equipment Scanner

Sacramento Army Depot

ES-33581

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 Production Acceptance Test and Evaluation (PATE). - When specified in the contract, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. - The shelter facility shall be constructed in accordance with the drawings listed on ES-33581 and the requirements specified herein, unless otherwise specified in the contract.

3.3 Electrical Installation/Performance (Shelter Facility). - (See 4.6)

3.3.1 Power system wiring and components. - The completed in-place power distribution system shall meet the following requirements:

3.3.1.1. Continuity. - Continuity shall exist between end terminations of all power wires and shall be less than 1 ohm, unless specified herein. (See 4.6.1.1.)

3.3.1.2 Dielectric breakdown and insulation resistance. - All AC power wires, cables and components shall withstand without flashver or insulation break-down, a potential of 500 volts DC between each of the ungrounded power line conductors and shelter facility ground applied continuously for not less than 30 sec. The insulation resistance shall not be less than 100 megohms. (See 4.6.1.2)

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3.3.1.3 DC Power wiring - Continuity shall exist between the circuit terminations as shown on the DC power schematic and the resistance shall be less than 0.5 ohms. (See 4.6.1.3)

3.3.2 Power distribution and operational functions. - Main power branch distribution and circuit switching functions thereof shall meet the operational requirements as shown in the power schematic drawing when tested in accordance with 4.6.2.

3.3.3 Signal system wiring and components. - The completed in-place signal system installation shall meet the following requirements:

3.3.3.1 Continuity - Continuity shall exist between end terminations of all signal wires and shall be less than 1 ohm, unless otherwise specified herein. (See 4.6.3 and 4.6.3.1)

3.3.3.2 Dielectric breakdown and insulation resistance. - All signal wires, cables and components shall withstand without flashover or insulation breakdown a potential of 500 volts DC applied continuously for not less than 5 seconds. The insulation resistance shall not be less than 100 megohms. (See 4.6.3.2).

3.3.4 Telephone and Intercom. - The telephone and intercom in the shelter facility shall be capable of providing satisfactory voice communications between the MPS or CIS and the shelter facility.

3.3.5 Shelter Alarms. - The temperature and air flow sensors shall properly provide the proper responses when stimulated by conditions of increased temperature or decreased air flow. Under normal temperature or air flow conditions no alarms shall be present.

3.3.6 ECU Operation. - The ECUs shall properly operate in all the various cooling, heating and fan positions.

3.4 Electrical performance, shelter assemblage. - (See 4.7)

3.4.1 Operational tests. - When the tests of 4.7 are performed, all circuits shall be operational.

3.5 Vehicular transportation.- The shelter facility shall be capable of being transported over cross-country terrain by military vehicle without sustaining any damage such as buckling, cracking, delamination, or other permanent deformation, and all components of the shelter facility shall remain secure. (See 3.17 and 4.8).

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3.6 Railroad transportation. - The shelter facility shall be capable of being loaded, blocked and braced on a flatcar for shipment by rail and shall be subjected to and withstand the tests of 4.9 without sustaining any permanent damage such as buckling, cracking, delamination or other permanent deformation, and all components of the shelter facility shall remain secure. (See 3.17 and 4.9)

3.7 Drop test. - The shelter facility shall be capable of withstanding drops of 18 inches onto concrete without sustaining any permanent damage such as buckling, cracking, delamination, or other permanent deformation, and all components of the shelter facility shall remain secure. (See 3.17 and 4.10)

3.8 Air tightness test. - The shelter facility shall show no evidence of air leakage through seams, joints, rivets, etc., or modified exterior surfaces when tested for tightness in accordance with 4.11.

3.9 Humidity. - The shelter facility shall be operable without degradation in specified performance, and shall sustain no physical damage during and after prolonged exposure to a relative humidity of 95 percent. (See 4.12)

3.10 Mounting inserts. - All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive and, when installed, each insert shall be capable of withstanding the tensile load and the torque load referenced in 4.13. There shall be no damage or permanent distortion to the shelter mounting surface or insert head or the threads and the inserts shall not have rotated from its original position. Insert threads shall be free of paint.

3.11 Acoustical Noise. - The acoustical noise shall not exceed the criteria specified in MIL-STD-1474(MI), Table 2, Category F. (See 4.14).

3.12 Cleaning. - (See 3.23)

3.12.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in applicable portions of MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

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3.12.2 Units. - After assembly, the equipment shall be cleaned thoroughly and shall be free from excess solder or flux, loose hardware and other foreign material.

3.13 Finish. - The shelter facility shall be finished in accordance with MIL-F-14072 and the equipment drawings. (See 3.23 and 4.4)

3.14 Marking. - (See 3.23 and 4.4)

3.14.1 General. - Marking shall conform to MIL-M-13231 and the drawings, unless otherwise specified.

3.14.2 Nameplates.- Shelter facility shall be provided with a nameplate in accordance with the drawings. Each nameplate shall be imprinted with a sequential serial number.

3.14.3 Aircraft loading data plate. - The shelter facility shall be provided with data plates in accordance with the drawings, imprinted with the dimension which locate the center-of-gravity of the shelter facility and the assemblage in the three planes; the gross weight of the shelter facility and assemblage; and the bearing pressure on the bottom of the skids. The location of the centers-of-gravity shall be determined by the contractor using a PATE Unit of production. (See 6.1)

3.14.4 Interior and exterior marking.- All interior and exterior markings shall be as defined on the drawings, unless otherwise specified.

3.15 Riveting. - Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces. Rivets shall conform to requirement 12 of MIL-STD-454. (See 3.23 and 4.17).

3.16 Seal, water vapor. - All joints of the framing for the power and signal entrance boxes and air conditioners shall be sealed to provide a barrier against the entrance water vapor to the core material of the shelter panels and to the interior of the shelter. All joints and edges shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws and fasteners shall be a one-part sealant Silastic, 738 as made by Dow Corning, or equal using the manufacturer's recommended primer. (See 4.4)

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3.17 Securing of Parts. - Brackets, lugs, flanges, inserts bolts, and other mounting arrangements shall retain components of the facility securely so that loosening, wear of mounting arrangements or permanent separation of parts or components will not occur when the equipment is subjected to the specified drop and transport tests. (See 3.23 and 4.17)

3.18 Soldering. - (See 3.23 and 4.17)

3.18.1 Solder and flux. - Solder and flux shall be as defined on the drawings, unless otherwise specified.

3.18.2 Acid or acid salts. - No acid or acid salts shall be used in preparation for or during soldering; however, exception permitted for preliminary tinning of electrical connections and for soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.18.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign materials remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.19 Tropicalization of Material.- Canvas, webbing, wood, etc., shall be treated to be resistant to moisture and fungi as indicated on the drawings as follows:

3.19.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

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3.19.2 Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.19.3 Flexibility. - Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.20 Welding. -

3.20.1 Welds. - Welded joints shall be as defined by the drawings unless otherwise specified and shall be such that grinding on the finished weld will be unnecessary except where specified on the drawings. Inert gas-shielded arc welding (MIG) using helium, argon, or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858 CL.B. and MIL-W-45205 CL.B as applicable. All surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used.

3.20.2 Process.- Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions and other harmful defects. Welded assemblies shall be cleaned to remove any scale oxidation products and excess flux. Any acid used in cleaning shall be completely neutralized and removed.

3.21 Wiring and cabling. - (See 3.23 and 4.17) Wiring and cabling shall be in accordance with the drawings unless otherwise specified and shall be neat and sturdy.

3.21.1 Slack. - Wires and cables shall be as short as practical except that sufficient slack shall be provided:

(a) To prevent undue stress on cable forms, wires, and connectors.

(b) To permit removal from rack of operational equipments, having wiring or cables to connectors at back of equipment, so that cable may be connected or disconnected.

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(c) To facilitate field repair of broken or cut wires.

(d) To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.21.2 Protection. - Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation materials.

3.21.3 Splicing. - Wires in a continuous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the drawings.

3.21.4 Solderless terminal lug connections.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of MIL-T-7928. (See 4.15)

3.21.5 Identification wiring.- Wiring shall be identified by color coding and wire pair marking.

3.21.6 Cord Assemblies. - Power cable and cord assemblies shall be fabricated in accordance with the drawing unless otherwise specified and meet the requirements and tests of MIL-C-3885. (See 4.4)

3.22 Interchangeability. - Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See Requirement 7 of MIL-STD-454) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.23 Workmanship. - (See 4.17) The shelter facility shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454.

3.24 EMI/EMC. - The OX-54/TYC-39(V) shall meet the following emission and susceptability requirements of MIL-STD-461A, Notice 4:

CE02	CS02	RE02	
CE03	CS06	RE02.1	(1) RS03
CE04			
CE05			

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(1) The entire shelter assemblage shall not exhibit degraded performance, malfunction, or undesirable response when immersed in electric fields of the following intensity using the modulation characteristics specified:

Freq. Range (HZ)	E-Field(v/M)	Modulation	Characteristic
10K-2M	1	AM, 50%	1KHz tone
2M-30M	20	AM, 50%	1KHz tone
30M-76M	50	FM, 8KHz Dev.,	1KHz tone
76M-400M	50	AM, 50%	1KHz tone
400M-2G	50	Pulse 0.1 microsec,	800 PPS
2G-12.4G	10	Pulse 0.1 microsec,	800 PPS

3.25 High temperature and ECU cooling.- When tested as specified in 4.20, the shelter assemblage shall meet the following requirements:

(a) Storage and Transit. 149°F (65°C).

(b) Operation: 120°F (48.9°C) external ambient temperature plus the effects of solar radiation at the rate of 360 BTU per square foot per hour. Ambient air temperature at the inlet ports of the ECUs shall not exceed 120°F (48.9°C). The shelter assemblage shall be capable of full operation at this temperature after all equipment has been operating for 30 minutes.

(c) ECU breakdown operation. Under abnormal or emergency conditions it shall be possible to obtain satisfactory equipment operation for a period of at least 2 hours under the external ambient temperature plus solar radiation defined above during the absence of one of the two environmental control units.

Individual inventory comsec equipments whose design specification do not state or meet the above requirements shall be exempt from these requirements.

3.26 Low temperature and ECU heating.- When tested as specified in 4.21, the shelter assemblage shall meet the following requirements:

(a) Storage and Transit: - 60°F (- 51.1°C)

(b) Operation: -50°F (-45.5°) external ambient temperature. The shelter assemblage shall be capable of providing full operation with an external ambient temperature of -50°F (-45.5°C) and a stabilized internal temperature of plus 32°F (0°C).

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Full operational capability shall be provided after all equipment has been operated for 2 hours. External heaters if required may be used to meet the transient warmup requirements. After complete start-up the equipment shall provide full operational capability for a minimum of six continuous hours without the use of environmental control facilities and with an external air ambient temperature of -25°F (-31.7°C).

Individual inventory comsec equipments whose design specifications do not state or meet the above requirements shall be exempt from these requirements.

3.27 Cable assemblies, power and audio frequency. - All power and signal cable and cord assemblies shall be fabricated in accordance with the drawings unless otherwise specified and meet the continuity, dielectric, insulation resistance and pull-out requirements of MIL-C-3885. (See 4.4)

3.28 TEMPEST - The shelter assemblage shall be capable of meeting the requirements of NACSEM 5100 for the following:

(a) Electric Radiation (ER)

(b) Powerline Conduction (PLC)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inpsections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. - All quality assurance operations required of the contractor shall be subject to Government verification at scheduled or unscheduled intervals. Verification will consist of the following:

(a) Surveillance of the contractor operations to determine that practices, methods, and procedures of the contractor's product assurance system are being applied, and are in fact, accomplishing the objectives of total compliance of the product with the requirements of this specification.

4.1.2 Accommodation and Assistance. - The Government quality assurance representative shall have the right to access any area

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of the contractor's or his subcontractor's premises where any part of the work is being performed. The Government quality assurance representative shall be afforded unrestricted opportunity to verify conformance of the product with specification requirements. The contractor shall make his inspection equipment available for use by the Government quality assurance representative for verification purposes. The contractor's personnel shall be made available for operation of such inspection equipment as required. All facilities and services necessary for the placement operation and maintenance of these test equipments shall be provided by the contractor.

4.2 Classification of Inspection. - Inspections shall be classified as follows:

(a) PATE Inspection (See 4.3). Does not include packaging.

(b) Inspections covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection of equipment before packaging. (See 4.5)

(d) Quality conformance inspection of packaging (See 4.18 and 4.19 if applicable).

4.3 PATE. - Unless otherwise specified in the contract or purchase order, the PATE inspection shall be performed by the contractor. (See 3.1)

4.3.1 PATE Units - The contractor shall furnish the number of PATE units as specified on the contract.

4.3.2 PATE Inspections. - The PATE inspections shall consist of inspections specified in Table I.

4.3.3 PATE Data. - The PATE test plan and test report(s) shall be required as in the contract or purchase order.

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TABLE I - PATE Inspection

Inspection		Rqmt Para	Insp Para
1.	Inspection covered by subsidiary documents		4.4
2.	Interchangeability	3.22	4.16
3.	Mounting Insert	3.10	4.13
4.	Solderless-terminal-lug	3.21.4	4.15
5.	Humidity	3.9	4.12
6.	Visual & Mechanical	3.23	4.17
7.	Electrical Installation/ Performance (Shelter Facility)	3.3	4.6
8.	Electrical Performance shelter assemblage	3.4	4.7
9.	Accoustical Noise	3.11	4.14
*10.	Vehicular transportation	3.5	4.8
*11.	Railroad transportation	3.6	4.9
*12.	Drop tests	3.7	4.10
13.	Airtightness	3.8	4.11
14.	High Temperature & ECU Cooling	3.25	4.20
15.	Low Temperature & ECU Heating	3.26	4.21
16.	EMI/EMC	3.24	4.23
17.	TEMPEST	3.28	4.22

* The asterisk labeled tests must all be done on the same sample in the order defined in the above table. After successful completion of the drop tests the test sample must be subjected and pass the airtightness test of 4.11. The contractor must perform at the end of each of the asterisk tests a V&M inspection per 4.17 and successfully pass such inspection. The contractor may elect to run only a final Electrical Inspection (See 4.6) at the end of the drop test in lieu of the requirements of 4.8, 4.9, which require such testing after the Vehicular transportation and Railroad tests. However, if this approach is taken and an Electrical test failure is uncovered in the final Electrical test (see 4.6) after drop tests are completed, the contractor must correct and run all asterisk tests in the order defined in the table and successfully pass the final Electrical test (4.6). Subsequently, the test sample must meet the airtightness requirements after being subjected to tests of 4.11.

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4.4 Inspection covered by Subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order:

ITEM	WHERE REQUIRED
Cable & Cord assemblies	3.27 & 3.21.6
Finish	3.13
Marking	3.14
Welds	3.20.1
Seal, Water vapor	3.16
Tropicalization of material	3.19

4.5 Quality conformance inspection of equipment before packaging. - The contractor shall perform the inspection specified in 4.4 and 4.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records as an element of Government verification. (See 4.1.1)

4.5.1 Group A Inspection. - Each shelter facility shall be inspected for conformance to the inspections specified in Table II. No failures shall be allowed. The interchangeability test shall be performed prior to all other Group A inspection.

4.5.1.1 Group A Non-Compliance. - If a unit fails Group A inspection, the unit shall be rejected. The contractor shall immediately investigate the cause of failure and shall report to the Government inspector and/or technical agency (if OX-54 is rejected)the results thereof and the details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector and/or technical agency does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of the failure, the matter shall be referred to the contracting officer for resolution.

4.5.1.2 Order of inspection within Group A. - Group A inspection shall be perfomed in the order listed in Table II.

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TABLE II - Group A Inspection

Inspection	Rqmt Para	Test Para	AQL %
Visual & Mechanical	3.23	4.17	No failures allowed
Electrical Installation/Shelter Performance	3.3	4.6	No failures allowed
Interchangeability	3.23	4.16	No failures allowed

4.5.2 Group C Inspection. - This inspection shall be as listed in Table III

TABLE III - Group C Inspection

Inspection	Rqmt Para	Insp Para
Subgroup I		
Electrical Performance/ Shelter assemblage	3.4	4.7
*Mounting Inserts	3.10	4.13
Airtightness	3.8	4.11
*This inspection shall be performed prior to Group A inspection and prior to installation of brackets, rack, etc.		
Subgroup II		
Solderless Terminal Lug connections	3.21	4.15
Subgroup III		
Accoustical noise	3.11	4.14
*Vehicular transportation	3.5	4.8
*Railroad transportation	3.6	4.9
*Drop tests	3.7	4.10
Humidity	3.9	4.12

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Subgroup III (continued)

High temperature & ECU cooling	3.25	4.20
Low temperature & ECU heating	3.26	4.21
Tempest	3.28	4.22
EMI/EMC	3.24	4.23

*The tests will be performed on the same unit in the order specified in the above Table. After successful completion of the drop tests the test sample must be subjected to and successfully pass the airtightness test of 4.11. The contractor must perform at the end of each of the asterisk tests a V&M inspection per 4.17 and successfully pass such inspections. The contractor may elect to run only a final Electrical inspection (See 4.6) at the end of the drop test in lieu of the requirements of 4.8 and 4.9 which require such testing after the vehicular and railroad tests. If this approach however is taken and an Electrical test failure is uncovered in the final Electrical test (See 4.6) after the drop tests are completed the contractor must correct and rerun all asterisk tests in the order defined in the Table and successfully pass the final Electrical test. (4.6). Subsequently, the test sample must meet the airtightness requirements after being subjected to the tests of 4.11.

4.5.2.1 Sampling for Group C. -

(a) Subgroup I.- One (1) unit shall be selected for testing from each 10 units or fraction thereof. In the case of mounting inserts four each size inserts shall be selected from the hardware associated with one (1) unit from each 10 shelter facilities or fraction thereof.

(b) Subgroup II. - Five specimens of each combination of wire and terminal produced each week shall be prepared for inspection. A specimen is a one (1) foot length of wire on which a terminal lug has been applied using the same tools and procedures used in crimping the terminal lugs in production.

(c) Subgroup III. - One (1) unit be selected from each 50 units or fraction thereof produced.

4.6 Electrical Installation/Performance (Shelter Facility) Test. The shelter facility shall be subjected to the tests of 4.6.1 thru 4.6.3 below. Before applying power to the shelter facility, an electrical ground connection at the power entrance box shall be made. Power to be supplied shall be 120/208 VAC, 3Ø, 60 Hertz. Complete checklists/data sheets shall be completed by the contractor for all electrical tests and shall be provided to the Government upon request.

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4.6.1 Power systems and components installation tests. - Refer to the Power Schematic.

a. Remove or disconnect the following electrical loads:

(1) All fluorescent, incandescent and neon lamps.

(2) All fluorescent lamp starters.

(3) Heaters, 28 VDC power supplies, air conditioners and intercom.

4.6.1.1 Continuity. - Continuity resistance tests which meet the requirements of 3.3.1.1 shall be performed on each power line circuit in accordance with the following paragraphs. A calibrated low resistance ohmmeter or equivalent shall be used in performing the test.

a. Place the following controls in the ON position:

(1) All circuit breakers.

(2) Fluorescent lamp switches.

(3) Incandescent lamp switch.

(4) Blackout bypass switch (to bypass position).

b. Measure continuity resistance as follows:

(1) From pin A of power input connector at the power entrance box to the black wire terminations of each fluorescent and incandescent lamp, air conditioner No. 1 and 2, convenience receptacles, intercom receptacle and AC power cables for each KG-13, KG-30 and SN-394 or AN/UYK-22.

(2) From pin B of power input connector at the power entrance box to the red wire termination of heater No. 1, air conditioners no. 1 and 2, DC power supply receptacle, and AC power cables for KWX-11 No. 1, 2. 3, and 4.

(3) From pin C of power input connector at the power entrance box to the blue wire termination of heater No. 2, air conditioners No. 1 and 2, and AC power cables for KWX-11, No. 5, 6, 7, 8 and 9.

(4) From pin N of power input connector at the power entrance box to the white wire termination of each fluorescent

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and incandescent lamp, each AC power receptacle and each equipment AC power cable.

(5) From pin G of power input connector at the power entrance box to Ground Terminal E1, the green wire termination at each AC power receptacle and each equipment AC power cable.

c. No continuity shall exist between each pin of the power entrance connector and each of the remaining pins.

4.6.1.2 Dielectric breakdown and insulation resistance. - The dielectric breakdown and insulation resistance of all wires, cables, and components shall meet the requirements of 3.3.1.2. A nondestructive DC dielectric tester shall be used in performing the tests. Test procedures shall be as follows:

CAUTION: Take preventative measures that will preclude personnel from coming into contact with conductors under test during performance of ALL dielectric and insulation resistance tests. Discharge each conductor to ground following each test in order to eliminate the danger of electric shock.

a. Retain all circuit controls tested under 4.6.1.1 above in the ON position except fluorescent lamp switches.

b. Disconnect at the meter switch all lines to the phase indicator circuitry. Isolate the disconnected lines so as not to short to each other or to shelter facility ground.

c. Test between pin A of the power input connector and all other pins of the power input connector strapped to shelter facility ground. Adjust tester for an output of 500 volts DC and maintain this potential for 30 seconds.

d. Repeat step c above for the following test points:

(1) Pin B of the power input connector with all other pins of the power input connector strapped to shelter facility ground.

(2) Pin C of the power input connector with all the pins of the power input connector strapped to shelter facility ground.

(3) Pin N of the power input connector with all other pins of the power input connector strapped to shelter facility ground.

4.6.1.3 DC power wiring. - The continuity requirement of 3.3.1.3 shall be met. Test procedures shall be as follows:

(a) Place all DC circuit breakers in the ON position and the voltmeter selector switch in the 28 V position.

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(b) Open DC monitor panel and DC power supply drawers as necessary.

(c) Test continuity between the 28 volt power supplies and voltmeter for each DC-DC converter and the air sensors as indicated on the DC power schematic. Test for shorts between other conductors and ground.

(d) Verify that no continuity exists between the 28 volt power supplies and the DC-DC converters and air sensors when the circuit breakers are placed in the off position.

(e) Place voltmeter selector switch in the +5V position. Test continuity between +5V DC-DC converter and the voltmeter and LCES as indicated on the DC power schematic. Test for shorts between other conductors and ground.

(f) Using the appropriate voltmeter selector switch settings repeat the test of step (e) above for each DC-DC converter and appropriate LCES or KWX-11 termination.

(g) Test the continuity of the remaining air sensor and associated LCES Interface wiring as indicated on the DC power schematic.

4.6.2 Power distribution and operational function tests. - Power distribution and operation function tests shall be performed on each power line circuit in accordance with the following paragraphs.

4.6.2.1 Test preparation. - (Refer to power schematic).

(a) Replace or reconnect all electrical loads and voltmeter which were removed or disconnected by 4.6.1.a and 4.6.1.b.

(b) Close power distribution box panel cover.

(c) Place all circuit breakers and switches in the OFF position.

(d) Provide a suitable 120 VAC test lamp with grounding plug or equivalent test indicator for the 120 VAC power receptacle.

(e) Close DC monitor panel.

4.6.2.2 Test for 120/208 volt AC prime power. -

(a) Apply 120/208V 3Ø, 60 hertz power to power input connector at the power entrance box. Power ON indicator lamps

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shall light. Correct phase indicator lamp shall light.

(b) Rotate meter switch to position L1, L2, and L3 respectively. The voltmeters shall indicate specific voltage (approximately 120 volts) in each instance. The frequency meter shall indicate 60 hertz in each instance.

(c) Place main circuit breaker in the ON position. The neon lamps above the main circuit breaker shall light.

4.6.2.3 Tests of circuit breakers and switches. -

a. Lights circuit breaker (CB2)

(1) Place circuit breaker CB2 in the ON position. Neon lamp above CB2 shall light.

(2) Place blackout bypass switch and fluorescent lamp switches in the ON position. All fluorescent lamps shall light.

(3) Place incandescent lamp switch in the ON position. All incandescent lamps shall light.

(4) Place CB2 inthe OFF position. All lamps mentioned insteps (1) thru (3) shall go out.

(5) Replace CB2 in the ON position to restore light.

(6) With shelter door open, place blackout bypass in the blackout position. All fluorescent and incandescent lamps shall go out.

(7) Close shelter door. All fluorescent and incandescent lamps shall light.

(8) Place fluorescent lamps switches in the OFF position. All fluorescent lamps shall go out.

(9) Place incandescent lamp switch in the OFF position. All incandescent lamps shall go out.

(10) Open shelter door and restore lights by placing blackout bypass and fluorescent lamp switches in the ON position.

b. Intercom and convenience circuit breaker. - (CB3)

(1) Place circuit breaker CB3 in the ON position. The neon lamp above circuit breaker CB3 and test lamps in the intercom and convenience 1 and 2 receptacles shall light.

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(2) Place circuit breaker CB3 in the OFF position. The lamps mentioned in step (1) shall go out.

c. Heater No. 1 Circuit Breaker (CB4) -

(1) Place circuit breaker CB4 inthe ON position. The neon lamp above tandem circuit breaker CB4 and test lamp in Heater No. 1 receptacle shall light.

(2) Place CB4 in OFF position. The lamps mentioned in step (1) shall go out.

d. Heater No. 2 Circuit Breaker(CB5)

(1) Repeat steps (1) and (2) of procedure c. Heater No. 1 circuit breaker (CB4) for CB5 and Heater No. 2.

e. Equipment Phase A Circuit breaker (CB6)

(1) Connect equipment AC power cables to appropriate jigs or fixtures as necessary to ensure safety.

(2) Place circuit breaker CB6 in the ON position. The neon lamp above circuit breaker CB6 shall light.

(3) For each Phase A equipment as indicated on the power schematic, measure AC voltage at the equipment AC power cable termination using a calibrated AC voltmeter (0-150 volt scale). The voltage reading shall be within $\pm$ 2% of the voltage measured in paragraph 4.6.2.2b.

(4) Place circuit breaker CB6 in the OFF position. The neon lamp above CB6 shall go out.

f. Equipment Phase B circuit breaker (CB7).

(1) Repeat steps (1) through (4)of procedure e. Equipment phase A circuit breaker (CB6) for CB7 and phase B equipment as indicated on the power schematic.

(2) Return CB7 to the ON position. Verify operation of fan in DC power supply rack.

(3) Place circuit breakers on the DC power supply rack for PS-1 and PS-2 28 volt power supplies in the ON position.

(4) Place circuit breakers CB 11 on the DC monitor panel in the ON position. The green light above CB 11 shall light and the red light above CB 12 shall light.

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(5) Place circuit breaker CB 11 in the OFF position. The lights on the DC monitor panel shall go out.

(6) Place circuit breaker CB 12 on the DC monitor panel in the ON position. The green light above CB-12 shall light and the red light above CB 11 shall light.

(7) Place all circuit breakers on the DC monitor panel in the ON position. The green light above CB 11 shall light and the red light above CB 11 shall go out; the green light above CB 12 shall remain lit.

(8) Rotate the voltmeter selector switch through all settings and monitor the voltmeter for correct voltage indications. Using a calibrated DC voltmeter, check for proper voltage readings at the respective LCES and KWX-11 equipment locations under the appropriate LCES and KWX-11 load conditions. Adjust DC-DC converters as required.

(9) PLace all circuit breakers on the DC monitor panel in the OFF position. The lights on the DC monitor panel shall go out.

(10) Place circuit breakers turned on by steps (2) and (3) above in the OFF position.

g. Equipment Phase C circuit breaker (CB8).

(1) Repeat steps (1) through (4) of procedure e. Equipment phase A circuit breaker (CB6) for CB8 and phase C equipments as indicated on the power schematic.

h. Air conditioner No. 1 circuit breaker (CB9)

(1) Place air conditioner No. 1 circuit breaker (CB9) in the ON position, neon lamp above CB9 shall light.

(2) Perform turn-on procedures for the air conditioner in accordance with its operator's manual. The air conditioner shall function properly.

(3) Turn air conditioner off, place CB9 in the OFF position, neon lamp above CB9 shall go out.

i. Air Conditioner No. 2 circuit breaker (CB 10)

(1) Repeat steps (1) through (3) of procedure H, air conditioner No. 1 (CB9) for CB 10 and air conditioner No. 2.

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4.6.2.4 Telephone and Intercom. - Verify telephone and intercom operation by connecting a compatible instrument to the appropriate pins of their respective signal entrance panel connectors and establishing voice communication.

4.6.2.5 Shelter Alarms . - Verify shelter door, air flow and temperature alarms by activating the alarms and checking for appropriate open and closed circuit conditions with a multimeter.

4.6.3 Signal system wiring and components installation. -

4.6.3.1 Continuity Test. - The point-to point continuity test of all signal lines shall meet the requirements of paragraph 3.3.3.1 as described in the following paragraphs. A low resistance continuity tester (Automatic or manual) or equivalent shall be used to test continuity resistance. The test setup shall include test cable connectors such that continuity measurements are made with connectors mated, (i.e., pin-to-pin continuity not allowed where termination through connectors is normally performed).

(a) Test continuity between each pin of each signal connector at the signal entrance panel and its respective equipment termination at the equipment rack as indicated on the signal schematic.

(b) Remove all signal connectors from the LCES.

(c) Test continuity between each pin of each LCES connector and its respective equipment termination as indicated on the signal schematic. (Resistance measurements for the SN-394 sync initiate circuit shall be as indicated on the signal schematic.

4.6.3.2 Dielectric breakdown and insulation resistance tests. - The dielectric breakdown and insulation resistance of all signal lines, cables, and components shall meet the requirements of 3.3.3.2. A nondestructive tester shall be used in performing the tests. Disconnect the LCES signal cables and remove all resistive loads. Test points are as follows: (See caution note under 4.6.1.2)

(a) Each conductor at the signal entrance box to all other conductors connected to ground.

(b) Each conductor at the LCES signal cables to all other conductors connected to ground.

4.6.4 ECU operation. - The ECUs shall be tested by examining their operation in the various heating/cooling and fan speed modes using the startup procedures outlined in the ECU manual. The ECUs shall properly operate in each mode examined.

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4.7 Electrical performance, shelter assemblage.- The following tests shall be performed with all components and operational equipments installed as indicated on the power and signal schematics. Complete checklists/data sheets shall be compiled by the contractor for the following tests and shall be provided to the Government upon request. (See 6.1)

4.7.1 COMSEC circuits. - Connect the two KG-13 circuits back-to-back on the black side at the CIS patch panel, insure that both COMSEC equipments have identical keying material. Verify traffic transmission in both directions at the applicable data rates. Repeat for the two KG-30 circuits and the KW-7 circuits two at a time.

4.7.1.2 Telephone and Intercom. - Verify telephone and intercom operation by establishing voice communication between the MPS or CIS and the OX-54/TYC-39(V). (See 3.3.4)

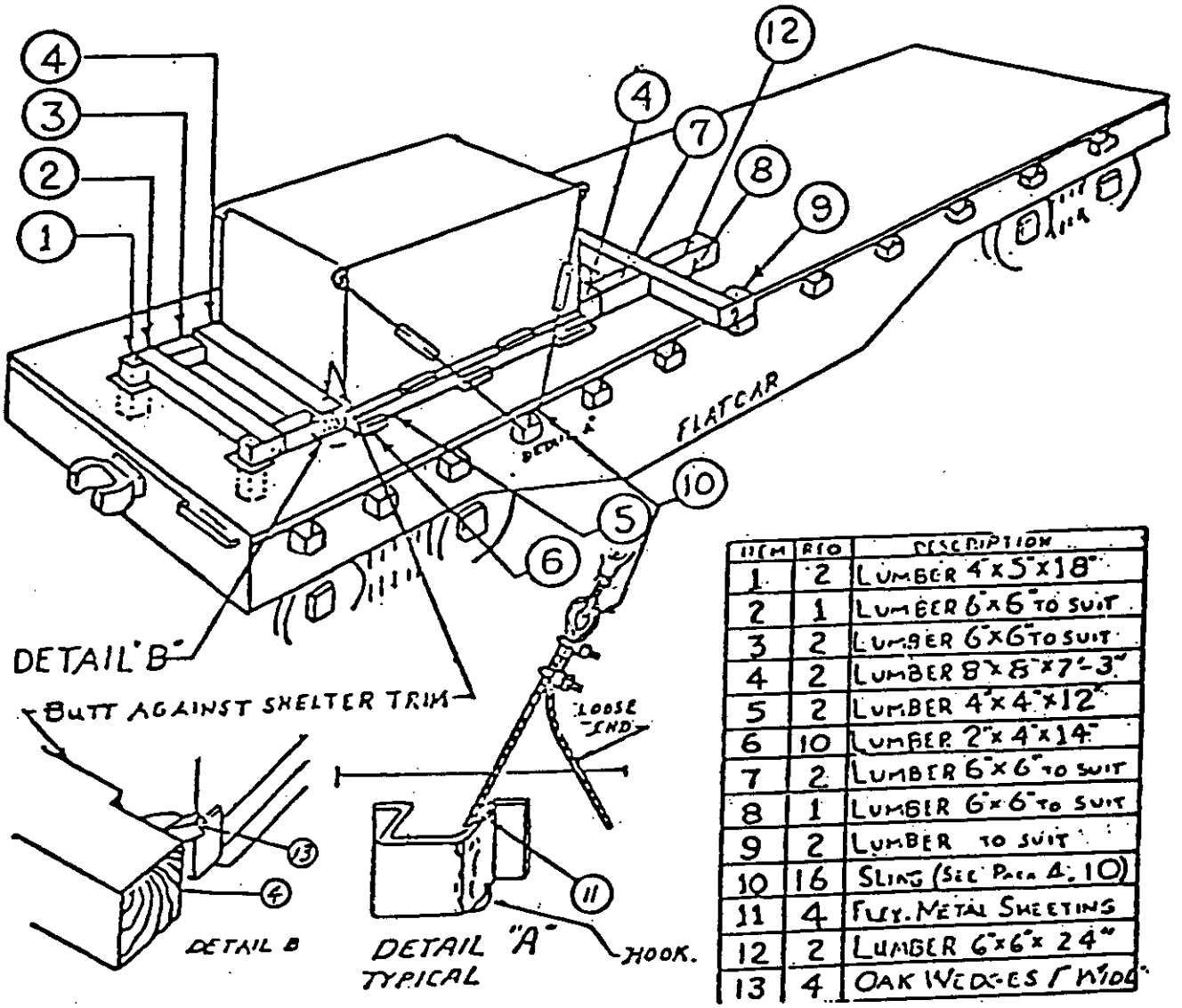
4.7.1.3 Shelter alarms. - Verify shelter door air flow and temperature alarms by activating the alarms at the OX-54/TYC-39(V) shelter and monitoring the shelter status at the MPS. (See 3.3.5)

4.8 Vehicular Transportation test. - The shelter facility with dummy loads to simulate operational equipment shall be secured (and blocked to prevent shifting) to a Government furnished 5 ton, M-813 truck and transported over the specially prepared course described in 4.8.1 for 10 laps at each of the speeds 5 mph, 10 mph, 15 mph, and 18-20 mph. A lap is defined as traversing the course in one direction. The assemblage shall meet the requirements of 3.5 after completion of the test. (See notes at the end of TABLE I and Subgroup 3 TABLE III.

4.8.1 Vehicular Transportation Course . - The course for military truck transportation test shall have twelve (12) 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with the 4-inch face fully above ground. The top edges shall hve 1-inch by 1-inch chamfers. The sixth and twelfth timbers shall be placed at 45 degrees to the travel, and all other timbers shall be placed perpendicular to the direction of travel. The timbers shall be anchored down securely by an effective means. The surface which the timbers are placed shall be hard and level.

4.9 Rail Transport, Impact Test.- The shelter facility with dummy loads to simulate operational equipment secured to a flatcar shall be impacted by a loaded gondola car. The test shall be performed on a straight and level stretch of track. The test flatcar with its timber blocking and shelter tie-down

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ITEM	REQ	DESCRIPTION
1	2	LUMBER 4"x5"x18"
2	1	LUMBER 6"x6" TO SUIT
3	2	LUMBER 6"x6" TO SUIT
4	2	LUMBER 8"x8"x7'-3"
5	2	LUMBER 4"x4"x12"
6	10	LUMBER 2"x4"x14"
7	2	LUMBER 6"x6" TO SUIT
8	1	LUMBER 6"x6" TO SUIT
9	2	LUMBER TO SUIT
10	16	SLING (SEE PARA 4.10)
11	4	FLEX. METAL SHEETING
12	2	LUMBER 6"x6"x 24"
13	4	OAK WEDGES (WOOD)

NOTES:

a. The blocking lumber shall be notched to accomodate towing eyes, and bolts. Wedge item 12 between towing corner forgin and item 4 and notch (Detail B).

b. The blocking lumber shall be rigidly fasted to the floor of the flatcar..

FIGURE 1

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slings, shall be as specified in Figure 1 and shall meet the provisions of the tests as specified in 4.9.1. The test flatcar (less the shelter) and the two empty flatcars coupled to it shall each weigh 45,000 to 55,000 pounds. The loaded gondola car shall meet the provisions of the test as specified in 4.9.2 and shall have a total weight of 165,000 pounds. The loaded gondola car totaling 165,000 pounds traveling at 9 miles per hour (or equivalent energy conditions acceptable to the Government Technical Representative) shall be impacted against the stationary loaded test car coupled to the two empty flatcars with brakes off. The test shall consist of four impacts. The first two impacts shall be with the door end of the shelter facing the loaded gondola car; the second two impacts shall be with the door end of the shelter facing the opposite direction from the loaded gondola car. The lashing, blocking and wedges holding the shelter in place shall be tight at the start of each run. If the blocking is torn loose or the lashing is broken during impact, the run shall be repeated. After each impact the shelter assemblage shall meet the requirements of 3.6. (See notes at the end of TABLE I and Subgroup 3 TABLE III).

4.9.1 Test flatcar. - The wooden planking of the test flatcar bed shall be in good condition. All 6-inch by 6-inch blockshall be secured to the bed of the flatcar using 10-inch spikes on 12-inch centers. The blocking along the door end and opposite end of the shelter shall fit tightly against the corner castings along the full length of the castings except as necessary to clear the towing shackles, this shall be accomplished by means of power sawed notches. The blocking along the other two sides of the shelter shall fit tightly against the corner castings, along the full length of the castings. There four blocking members shall be notched or shimmed to contact the trim along the sides of the shelter. The 4-inch by 4-inch and 2-inch by 4-inch blocking shall be nailed to the bed of the flatcar as shown in the figure. Lashing shall consist of four wire rope slings, each including 3/8 inch diameter, 6-inch by 19-inch, flexible steel, preformed wire rope (having a breaking strength of 10,600 pounds minimum), turnbuckles, two hooks, with thimbles and fasteners as required; the breaking strength of the sling shall be 9000 pounds minimum. Before each impact, each sling shall be taut (600-800 pounds tension).

4.9.2 Loaded Gondola Car. - The load in the loaded gondola car shall be one of the following:

(a) A course size of either coal, crushed stone, gravel, concrete or metal scrap (other than sheet metal, metal shavings or other springy forms of metal).

(b) Chunk, slab, block or plate forms of the preceding acceptable materials, constrained to prevent shifting.

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4.10 Drop tests.-

4.10.1 Flat drop.- The shelter facility with dummy loads to simulate operational equipment shall be lifted 18 inches from the ground and allowed to fall freely with the skids impacting onto a hard substantial concrete surface. The test shall be performed once. Upon completion of this drop, the assemblage shall meet the requirements of 3.7.

4.10.2 Rotational drop.- The shelter facility with dummy loads to simulate operational equipment shall have one skid (or the knuckles of the skids) resting on a nominal 4-inch high timber. The outer edge of the opposite skid (or the opposite knuckles of the skids) shall be raised 18 inches and the shelter shall be allowed to fall freely onto a hard substantial concrete surface. This impact shall be performed four times, once for each bottom edge of the shelter. On completion of these drops the shelter assemblage shall meet the requirements of 3.7 when tested afterwards to 4.6 and 4.17.

4.11 Air tightness test.- The shelter facility when subjected to the following test, shall meet the requirements of 3.8.

(a) The air tightness shall be tested by introducing clean dry air into the closed shelter through an existing appropriate opening (i.e., drain fixture) to create an internal air pressure that exceeds the external by a minimum of 6 inches of water.

(b) A manometer graduated 0.1 inches or less shall be utilized to measure this differential pressure.

(c) The test apparatus shall include an air supply sufficient to maintain the required air pressure, an air flow regulator an an input air pressure gage.

(d) The shelter facility shall be tested for air tightness by painting the sealed joints, seams, rivets, etc., of the external modifications with Leak-Tac Formula No. 372 (American Gas and Chemicals, Inc., Leak-Tac Division) or equal soap and water solution.

(e) To minimize air loss, temporary seals, shims and tape are permitted to seal the shelter door, and air conditioner port opening not being tested.

(f) There shall be no leakage of the areas being tested as evidenced by bubbling or spurting of the test solution.

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4.12 Humidity Test. - The shelter assemblage shall be subjected to Procedure II, Method 507 of MIL-STD-810C. All vents, louvres, doors, and covers shall be open in Steps 1-3 inclusive. In Step 4, the door shall be closed (vents, louvres and covers still open) and the ECU and electronics energized. When the equipment has stabilized sufficiently to allow proper operation the operational test of 4.7 shall be conducted. In Step 5, the equipment/ECU shall be de-energized and the shelter door opened. During last five hours of each cycle in Step 6, the ECU/electronics shall be energized with door closed and when the equipment has been stablized sufficiently for operation the operational test of 4.7 shall be conducted. In Step 7, the doors, louvres, vents and covers shall be opened. In Step 8, the door shall be closed and the ECU/electronics shall be energized. The operational test of 4.7 shall be performed; in addition the shelter facility shall be inspected for physical damage resulting from the humidity (See 3.9).

4.13 Mounting Inserts. - Four (4) of each size insert shall be randomly selected for test. Each insert shall be tested with an axial load followed by a torque load in accordance with TABLE IV. After test the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample equipment. This test shall be conducted prior to installation of brackets, racks, etc.

TABLE IV - Mounting Insert Loads

Insert	Load Axial (lbs)	Torque (lbs)
5/16-18	2000	100
1/4-20	1300	60
#10-32	600	23

4.14 Acoustical noise test. - The shelter facility shall be located in an area remote from excessive machinery, vehicle and other noise operating sources to reduce ambient noise to a minimum, with the shelter door closed. Only necessary test personnel shall be present in the shelter. Using Octave Band analyzer, General Radio Corporation, Model 1558A, or equal, and Microphone, Piezoelectric, General Radio Corporation Model 1560-

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P6 or equal, record ambient noise levels over the frequency ranges specified in 3.11. Operate the exhaust blowers and repeat noise level recording at the same frequency levels as above. The noise levels, corrected from ambient shall meet the requirements of 3.11.

4.15 Testing of solderless terminal-lug connections.- Connections made with solderless terminal-lug shall be tested for conformance to 3.21.4.

4.16 Inspection for interchangeability. - The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.22. When a dimension is not within specified or design limits, it shall be considered a defect.

(a) Size and mounting dimensions of brackets and mounting plates.

(b) Size and mounting dimensions of racks and cabinets.

In addition, like removable assemblies (subassemblies), units (PC Cards, etc.) such as power supplies, LCES Cards, etc., shall be replaced during this inspection with identical items to determine if they are physically interchangeable. Verification of functional interchangeability shall be done during the tests of 4.6 and should consist of verification that required responses in terms of voltages, etc., are received with another identical replaceable unit assembly/subassembly.

4.17 Visual/Mechanical Inspection. - Visual and mechanical inspection shall be in accordance with MIL-STD-252 and the following inspections: parts, mechanical construction, loose hardware (i.e., brackets, flanges, lugs, inserts, bolts,and other mounting arrangements, etc) application of specified treatments and miscellaneous defects. The equipment and controls shall be inspected for proper mechanical and electrical operation. Wiring, soldering, ground, riveting, welding, cabling, cleaning, and finishes, etc., shall be inspected for workmanship.

4.18 Quality conformance inspection of packaging. - Quality conformance inspection of packaging shall be in accordance with Group A, Group B and, when required (see 6.2.e), the Group C (rough handling) requirements of MIL-P-116. Inspection lots shall be in accordance with MIL-P-116.

4.19 Rough handling test (packaging). - When rough handling test in accordance with MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operation malfunction caused by rough handling:

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Paragraph 4.6 Power and Signal wiring

Paragraph 4.17 Visual and Mechanical

4.20 High Temperature and ECU Cooling (3.25). - The shelter assemblage shall be subjected to Test Method 501, Procedure II of MIL-STD-810C. The internal chamber temperature shall be 149°F (65°C) in Step 4 and 120°F (48.9°C) plus the effects of solar radiation at the rate of 360 BTU/sq ft/hr in Step 7. In Steps 1 through 6 inclusive all shelter doors, louvres, vents, and covers shall be closed. In step 8 open all louvres, vents, and covers and close shelter door. Turn on all electronics and ECUs and verify that after 30 minutes all equipment is operational by performing the operational test of 4.7. After this inspection, shutdown one ECU and verify that the assemblage is fully operational after 2 hours of ECU shutdown by performing the operational test of 4.7. Subsequently, the shelter assemblage shall be powered down and returned to ambient conditions (Step 9) In Step 10, perform another operational test of 4.7 to verify satisfactory operation. Inventory COMSEC whose design specification do not state or do not meet the above requirements are exempt from these requirements. The test plan shall clearly define if any such equipment are incapable of meeting these requirements.

4.21 Low Temperature and ECU Heating. - (3.26) The shelter assemblage shall be subjected to the test of Method 502, Procedure I of MIL-STD-810C. Step 2 storage temperature shall be -60°F (-51.1° C) and operation temperature of -50°C (45.5°) in Step 4. In Steps 1-4 inclusive all louvres, covers, vents, and doors shall be closed. In Step 5 all louvres, vents, covers shall be opened and the shelter door closed. At this point, the ECU and electronics shall be turned on and the shelter assemblage shall reach full operational capability with 2 hours with an internal shelter ambient of 32°F. This will be verified by performing the operational test of 4.7. External heaters may be used to meet the transient warmup requirement. The chamber temperature shall then be raised to -25°F and the shelter assemblage shall after temperature stabilization run for 6 continuous hours without the use of ECUs. Verification of successful operation under these conditions will be performed by running the operational test of 4.7 near the end of the 6 hours period. In Step 7, perform another operational test per 4.7 to verify compliance to operational requirements. Inventory COMSEC whose design specifications do not meet the above requirements are exempt from these requirements the test plan shall clearly define if any such equipment are incapable of meeting these requirements.

4.22 TEMPEST . - The shelter assemblage shall be set up in conjunction with communications interface and message processing shelters of the AN/TYC-39 system (see Figure 2). The shelter assemblage shall meet the Electric Radiation limits of NACSEM 5100 and its power lines shall meet the Black Line conduction limits of NACSEM 5100. Measurement equipment and test methods shall conform to NACSEM 5100 for ER and PLC tests.

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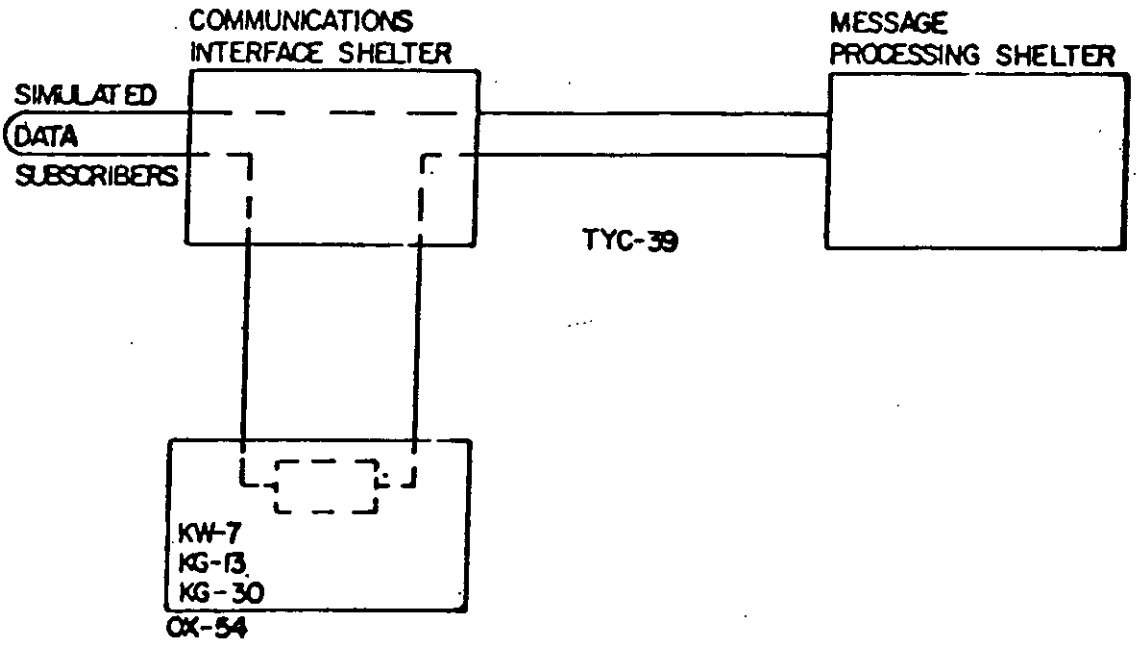


FIGURE 2 OX-54 Test Setup

4.23 EMI/EMC.- The shelter assemblage shall be tested for compliance with 3.24 using the approved measurement techniques of MIL-STD-462, NOTICE 3, as implemented by a contractor prepared, Government approved EMI Test plan.

5. PACKAGING

5.1 Packaging requirements. - Packaging requirements shall be in accordance with MIL-S-55507.

6. NOTES

6.1 Intended use. - The major operational equipments to be used in the OX-54/TYC-39(V) are as follows:

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2 each Electronic Key Generator KG-13
2 each Electronic Key Generator KG-30
2 each Electrical Synchronizer SN-394 or 1 each Crypto Auxiliary Set AN/UYK-22

18 each Electronic Tactical TTY Security Equipment KW-7.

9 each Fixed Plant Adaptor Unit KWX-11
1 each Line Conditioning Equipment Scanner (SM-B-814704)

(The shelter facility plus operational equipments shall constitute the shelter assemblage)

6.2 Ordering data . - Procurement documents should specify the following:

(a) Title, number and date of this specification and any admendment thereto.

(b) PATE and Group C inspection of the shelter assemblage requires use of contractor fabricated dummy loads or the Government furnished live loads (operational equipment) as specified in the bid request and contract.

(c) A list of Government furnished equipments required as part of the OX-54/TYC-39(V).

(d) A list of equipments to be Government loaded for testing purposes.

(e) When rough handling test is required (See section 4).

6.3 Inspection. - Inspection is the examination or testing or both supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination. - Examination consists of simple, generally nondestructive determinations of compliance, without use of special test equipment.

6.3.2 Testing. - Testing consists of determination of compliance, using technical means.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.4)

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6.5 Verification inspection. - Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.6 Tropicalization of material. - All materials of construction are specified on the drawings and are Fungus-inert. Substitution of any materials other than specified must be approved by the contracting officer. The material will only be approved after the contractor tests the material for Fungus-Resistance satisfactorily in accordance with requirement 508 of MIL-STD-810C

Custodian:
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Preparing Activity
ARMY-CR
Project No. 5895-A292

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No. 22-R255

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MIL-C-49246 (CR) Coder-Decoder Group OX-54/TYC-39(V)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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